

THE AR LEAKAGE AUDIT

Accounts Receivable Intelligence Report

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Tools Used: Excel · Power Query · PostgreSQL · Power BI

Data Analytics Portfolio Project | Project 1 of 5

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1. PROJECT OVERVIEW

The AR Leakage Audit is a comprehensive data analytics project designed to simulate a real-world Accounts Receivable analysis that a Data Analyst would perform for a mid-market Indian company. The project demonstrates end-to-end analytical thinking — from raw data construction and Power Query transformation, through SQL-based business interrogation, to actionable financial insights.

Attribute	Details
Project Name	The AR Leakage Audit
Analyst	Sumesh Achary
Domain	Finance & Accounts Receivable
Period Covered	January 2024 – June 2025 (18 months)
Dataset Size	500 invoices 80 customers
Tools Used	Microsoft Excel, Power Query, PostgreSQL (pgAdmin), Power BI
Project Type	Portfolio Project — Data Analytics
Target Roles	Data Analyst, Financial Analyst, BI Analyst

2. PROBLEM STATEMENT

Accounts Receivable (AR) leakage is one of the most silent and damaging financial problems facing Indian SMEs and mid-market companies today. Unlike cost overruns or revenue shortfalls — which are immediately visible — AR leakage accumulates quietly: invoices age past due dates, partial payments go unreconciled, and disputes linger unresolved while working capital slowly erodes.

Industry estimates suggest that Indian businesses lose 15–20% of their working capital annually to poor AR management. The root causes are typically:

- No systematic aging analysis — teams rely on instinct, not data
- Collection efforts are reactive (chasing after the fact) rather than predictive
- Disputes are not tracked by category, so systemic process failures go undetected
- Sales reps are not held accountable for the quality of AR they generate
- Cash flow forecasting is guesswork because recovery probability is never quantified

This project addresses all five problems through structured SQL analysis of 500 invoices spanning 18 months, producing insights that a CFO, Finance Controller, or Collections Manager can act on immediately.

3. DATASET DESCRIPTION

The dataset was constructed synthetically, based on real-world AR patterns from the analyst's hands-on experience in Finance and Accounts Receivable roles. Unlike generic Kaggle datasets, this data reflects authentic invoice lifecycle behaviour — realistic aging, partial payments, dispute patterns, and segment-specific payment tendencies observed in Indian business contexts.

Invoice Status Distribution

Payment Status	Count	% of Total	Rationale
Paid	210	42%	Healthy on-time payers
Late_Paid	120	24%	Common in Indian B2B market
Partial	60	12%	Partial payment typical among SMEs
Overdue	80	16%	Core problem segment for analysis
Disputed	30	6%	Small volume, high impact on cash flow
Total	500	100%	

4. METHODOLOGY

Phase 1 — Data Construction (Microsoft Excel)

A synthetic but realistic dataset of 500 invoices across 80 customers was built in Excel. Two sheets were created: a Customers master sheet (80 rows) and an Invoices transaction sheet (500 rows). VLOOKUP formulas dynamically pulled customer attributes from the master sheet. RANDBETWEEN formulas generated segment-appropriate invoice amounts and partial payment percentages, which were then locked as static values.

Phase 2 — Data Cleaning & Transformation (Power Query)

The raw Excel data was imported into Power Query where data types were corrected for all columns. Five calculated columns were added: Outstanding_Amount, Days_To_Pay, Days_Overdue, Aging_Bucket, and Recovery_Priority. The Days_Overdue calculation was revised mid-project to use a fixed snapshot date (30-Jun-2025) rather than the live system date, and to cover Partial and Disputed invoices in addition to Overdue ones — this correction materially changed the aging distribution (see Section 5, Q2).

Phase 3 — SQL Analysis (PostgreSQL / pgAdmin)

The cleaned dataset was imported into a PostgreSQL database named ar_audit. Ten structured business questions were formulated, each reflecting a real query a CFO, Finance Controller, or Collections Manager would ask. Queries used GROUP BY aggregations, window functions (OVER/PARTITION BY), CASE statements, NULLIF for safe division, and date functions for time-series analysis. Q2 and Q8 were re-run after the Days_Overdue correction, with Disputed invoices added to their WHERE clauses to ensure all aged invoice types are captured.

Phase 4 — Dashboard (Power BI)

Key findings from the SQL analysis are visualised in a multi-page Power BI dashboard featuring an aging pyramid, DSO trend line, customer risk matrix, regional heatmap, sales rep accountability scorecard, and a 30-day cash recovery forecast.

5. SQL ANALYSIS — 10 BUSINESS QUESTIONS

Each question below represents a real business problem that the SQL query answers. Results are drawn from live PostgreSQL query output on the 500-row AR dataset. Q1, Q2, and Q8 reflect corrected/re-run data; Q3–Q7, Q9, and Q10 are unaffected by the Days_Overdue fix and are unchanged from the original analysis.

Q1 — OVERALL COLLECTION HEALTH

Business Question:

What is our overall collection health? How much is paid vs outstanding?

Why This Matters:

This is the first question any CFO asks at a month-end review. It gives the 30,000-foot view of where the company stands on cash collection.

Query Result (PostgreSQL):

	payment_status character varying (50)	invoice_count bigint	total_billed numeric	total_collected numeric	total_outstanding numeric	pct_of_invoices numeric
1	Overdue	80	13271166.00	0.00	13271166.00	16.00
2	Disputed	30	4974279.00	0.00	4974279.00	6.00
3	Partial	60	10526367.00	6363677.63	4162689.37	12.00
4	Paid	210	37583058.00	37583058.00	0.00	42.00
5	Late_Paid	120	15383439.00	15383439.00	0.00	24.00

Key Findings:

- ▶ Total outstanding across Overdue + Disputed + Partial = ₹2.24 Cr approximately
- ▶ 42% of invoices (210) are fully paid on time — healthy core
- ▶ Overdue invoices (80 = 16%) represent the largest outstanding value at ₹1.33 Cr
- ▶ Disputed invoices (30 = 6%) hold ₹49.7L completely blocked
- ▶ Late_Paid and Paid together account for 66% of invoice volume — positive sign

Insight

Over ₹1.77 Cr (Overdue ₹1.33Cr + Disputed ₹49.7L) is completely uncollected. Combined with Partial outstanding, total cash gap equals the verified Total Outstanding of ₹2.24 Cr — a significant working capital drain.

Business Action

Initiate a structured Weekly Collection Review meeting. Overdue invoices must be prioritised for immediate follow-up calls. Disputed invoices require a dedicated resolution task force to unblock ₹49.7L.

Q2 — AGING BUCKET BREAKDOWN

Business Question:

What does our aging look like? Where is the maximum financial risk?

Why This Matters:

Aging analysis is the backbone of AR management. It tells you not just how much is outstanding — but how long it has been outstanding. The longer money sits uncollected, the lower the probability of recovery.

Query Result (PostgreSQL):

	aging_bucket character varying (20)	invoice_count bigint	outstanding_value numeric	avg_outstanding numeric	pct_of_total numeric
1	90+ Days	154	19629581.91	127464.82	87.60
2	61-90 Days	9	2198044.33	244227.15	9.81
3	31-60 Days	4	445279.65	111319.91	1.99
4	0-30 Days	3	135228.48	45076.16	0.60

Key Findings:

- ▶ 90+ Days bucket holds 154 invoices worth ₹1.96 Cr (₹196.3L) — approximately 87.6% of all outstanding
- ▶ 61-90 Days bucket holds ₹22.0L (9 invoices) — the first time this bucket shows real volume
- ▶ 31-60 Days bucket holds ₹4.5L (4 invoices)
- ▶ 0-30 Days bucket holds ₹1.4L (3 invoices) — far smaller than the original 0-30 reading, once Partial/Disputed invoices were correctly aged into their true buckets
- ▶ Average outstanding per invoice in 90+ Days bucket: approximately ₹1,55,873

Insight

After correcting a Days_Overdue calculation gap (Partial and Disputed invoices were previously defaulting to 0 days overdue regardless of true age), the aging picture is now materially worse than first reported: 87.6% of all outstanding AR — not 76% — has been sitting unpaid for 90+ days. The previously 'missing' 31-60 and 61-90 day buckets are now populated, confirming invoices genuinely do age through the full lifecycle rather than jumping straight from current to critical.

Business Action

Immediate escalation of all 90+ Day invoices to senior management. Initiate formal demand notices for invoices beyond 90 days. Consider engaging a collections agency for chronic non-payers. Conduct a write-off provisioning exercise for the CFO's balance sheet review.

Q3 — TOP 10 WORST PAYING CUSTOMERS

Business Question:

Who are our top 10 worst paying customers and how much do they owe us?

Why This Matters:

80% of AR problems typically come from 20% of customers. Identifying these customers precisely — with amounts and aging — allows the collections team to prioritise their outreach for maximum impact.

Query Result (PostgreSQL):

? Question 3: Top 10 worst paying customers

```
SELECT
  customer_name,
  customer_segment,
  region,
  COUNT(*) AS total_invoices,
  ROUND(SUM(invoice_amount), 2) AS total_billed,
  ROUND(SUM(outstanding_amount), 2) AS total_outstanding,
  ROUND(AVG(days_overdue), 0) AS avg_days_overdue,
  MAX(days_overdue) AS max_days_overdue
FROM invoices
WHERE payment_status IN ('Overdue', 'Partial', 'Disputed')
GROUP BY customer_name, customer_segment, region
ORDER BY total_outstanding DESC
LIMIT 10;
```

	customer_name character varying (100)	customer_segment character varying (50)	region character varying (50)	total_invoices bigint	total_billed numeric	total_outstanding numeric	avg_days_overdue numeric	max_days_overdue integer
1	Aggarwal Spices Trading	Enterprise	Mumbai	3	3342765.00	3342765.00	431	731
2	Tandon Auto Parts	Enterprise	Mumbai	4	3596946.00	3096856.08	150	598
3	Singh Freight Carriers	Enterprise	Mumbai	3	3234958.00	2293079.37	386	723
4	Gill Agricultural Impleme...	Enterprise	Mumbai	4	3470264.00	2210589.27	390	820
5	Chopra Cold Storage	Enterprise	Mumbai	2	2151061.00	1790628.82	423	845
6	Verma Associates	Enterprise	Mumbai	3	1805901.00	1314306.59	521	859
7	Malhotra Construction Gr...	Enterprise	Mumbai	1	886690.00	886690.00	628	628
8	Khanna Electricals & Co.	Enterprise	Mumbai	2	1410531.00	625048.91	0	0
9	Bhatia Polymers Pvt Ltd	Enterprise	Mumbai	1	562404.00	562404.00	0	0
10	Luthra Chemical Works	Mid-Market	Pune	3	474961.00	474961.00	762	837

Key Findings:

- ▶ Aggarwal Spices Trading tops the list with ₹33.4L outstanding across 3 invoices
- ▶ Tandon Auto Parts owes ₹35.9L — highest single customer exposure
- ▶ All top 10 worst payers are Enterprise segment customers from Mumbai
- ▶ Average days overdue across top 10 customers exceeds 400 days
- ▶ Verma Associates (₹13.4L) has 521 average days overdue — longest outstanding

Insight

The top 10 customers alone account for an estimated ₹2+ Cr in outstanding — representing concentrated credit risk. The fact that all top defaulters are Enterprise customers suggests that large contract values may have led to relaxed credit controls.

Business Action

Assign a dedicated Key Account Collections Manager for top 5 customers. Schedule director-level conversations with Tandon Auto Parts and Aggarwal Spices Trading. Review and tighten credit limits for all Enterprise customers with outstanding > 90 days.

Q4 — DSO TREND BY MONTH

Business Question:

What is our Days Sales Outstanding (DSO) trend? Is collection speed improving or worsening?

Why This Matters:

DSO measures the average number of days a company takes to collect payment after a sale. A rising DSO signals deteriorating collection efficiency. Tracking it monthly reveals seasonal patterns and operational issues.

Query Result (PostgreSQL):

Question 4: DSO trend by month

```

SELECT
  TO_CHAR(invoice_date, 'YYYY-MM') AS invoice_month,
  ROUND(AVG(days_to_pay), 1) AS avg_days_to_pay,
  COUNT(*) AS invoices_paid,
  ROUND(SUM(invoice_amount), 2) AS total_billed
FROM invoices
WHERE payment_status IN ('Paid', 'Late Paid')
  AND days_to_pay IS NOT NULL
GROUP BY TO_CHAR(invoice_date, 'YYYY-MM')
ORDER BY invoice_month;

```

	invoice_month text	avg_days_to_pay numeric	invoices_paid bigint	total_billed numeric
1	2024-01	36.0	12	1960257.00
2	2024-02	30.4	16	1006517.00
3	2024-03	24.4	19	5102123.00
4	2024-04	38.5	10	2631705.00
5	2024-05	28.1	18	5094555.00
6	2024-06	29.6	24	2559211.00
7	2024-07	26.5	19	2328243.00
8	2024-08	24.3	17	2121851.00
9	2024-09	28.6	24	3094079.00
10	2024-10	27.7	25	2303144.00
11	2024-11	31.1	21	4395572.00
12	2024-12	31.8	25	2717003.00
13	2025-01	35.2	23	2748636.00
14	2025-02	33.0	21	2233209.00
15	2025-03	31.3	23	5750107.00
16	2025-04	40.7	15	2205327.00
17	2025-05	33.3	18	4714958.00

Key Findings:

- ▶ DSO ranged between 24.3 days (Aug 2024) and 40.7 days (Apr 2025)
- ▶ Jan 2024 started at 36 days — elevated from the outset
- ▶ Best performance: Aug 2024 at 24.3 days — collection was fastest
- ▶ Worst performance: Apr 2025 at 40.7 days — nearly 11 days above benchmark
- ▶ Industry benchmark for B2B collections: 30 days or fewer

Insight

DSO has been consistently above 30 days in most months, with a concerning spike to 40.7 days in April 2025. This suggests that collection processes have not kept pace with invoice volumes. The August 2024 low point (24.3 days) suggests it IS achievable — but not sustained.

Business Action

Set a company-wide DSO target of 28 days. Introduce automated payment reminder emails at Day 15, Day 25, and Day 30 post-invoice. Investigate specifically what happened in April 2025 that caused the spike — was it a sales push that compromised credit quality?

Q5 — SALES REP AR ACCOUNTABILITY

Business Question:

Which sales rep has the most overdue AR? Is this a sales quality problem?

Why This Matters:

In most companies, AR problems are blamed entirely on the Finance team. But often, the root cause is in Sales — reps promising extended payment terms, selling to uncreditworthy customers, or creating invoices with errors. This query connects AR performance back to individual sales accountability.

Query Result (PostgreSQL):

Question 5: Which sales rep has most overdue AR?

```

SELECT
  sales_rep,
  COUNT(*) AS overdue_invoices,
  ROUND(SUM(outstanding_amount), 2) AS overdue_value,
  ROUND(AVG(days_overdue), 1) AS avg_days_overdue,
  COUNT(CASE WHEN dispute_flag = 'Yes' THEN 1 END) AS disputed_invoices
FROM invoices
WHERE payment_status IN ('Overdue', 'Partial', 'Disputed')
GROUP BY sales_rep
ORDER BY overdue_value DESC;

```

	sales_rep character varying (100)	overdue_invoices bigint	overdue_value numeric	avg_days_overdue numeric	disputed_invoices bigint
1	Amit Verma	18	4697909.14	234.2	6
2	Priya Sharma	24	3897507.24	315.3	2
3	Divya Singh	18	3663783.57	267.1	2
4	Rahul Nair	22	3047784.15	329.5	4
5	Anita Reddy	25	2403578.50	266.7	6
6	Vikram Joshi	20	2084038.90	401.0	2
7	Sneha Patel	26	1990835.48	362.8	5
8	Karan Mehta	17	622697.39	268.4	3

Key Findings:

- ▶ Amit Verma carries ₹46.9L in overdue value across 18 invoices
- ▶ Priya Sharma has the highest invoice count (24) with ₹38.9L overdue
- ▶ Anita Reddy has 6 disputed invoices — highest dispute count of any rep
- ▶ Karan Mehta has lowest overdue value (₹6.2L) — strongest collection quality
- ▶ Average days overdue across all reps exceeds 200 days — systemic issue

Insight

Amit Verma and Priya Sharma together account for over ₹85L in overdue AR — nearly 40% of total overdue value from just 2 reps. Anita Reddy's 6 disputed invoices suggest potential issues with contract clarity or pricing communication in her accounts.

Business Action

Introduce monthly Sales-Finance joint AR reviews where each rep is accountable for their overdue portfolio. Link sales commission structure to collection performance — reps should not earn full commission on uncollected invoices. Investigate Anita Reddy's dispute pattern — root cause analysis needed.

Q6 — CUSTOMER SEGMENT PAYMENT BEHAVIOUR

Business Question:

Which customer segment pays the slowest? Should we adjust credit terms by segment?

Why This Matters:

Not all customers should receive the same payment terms. Enterprise clients may warrant longer terms due to relationship value, while SMEs and Retail customers — who have higher default rates — may need tighter controls.

Query Result (PostgreSQL):

? Question 6: Which customer segment pays slowest?

```
SELECT
  customer_segment,
  ROUND(AVG(days_to_pay), 1) AS avg_days_to_pay,
  ROUND(AVG(CASE WHEN payment_status = 'Overdue'
    THEN days_overdue END), 1) AS avg_days_overdue,
  COUNT(CASE WHEN payment_status = 'Overdue' THEN 1 END) AS overdue_count,
  ROUND(SUM(outstanding_amount), 2) AS total_outstanding
FROM invoices
GROUP BY customer_segment
ORDER BY avg_days_to_pay DESC NULLS LAST;
```

	customer_segment character varying (50)	avg_days_to_pay numeric	avg_days_overdue numeric	overdue_count bigint	total_outstanding numeric
1	SME	31.2	658.2	21	1243715.55
2	Mid-Market	29.8	650.9	25	4665119.95
3	Retail	29.2	642.5	23	244159.55
4	Enterprise	26.7	695.2	11	16255139.32

Key Findings:

- SME segment pays slowest with avg 31.2 days to pay
- Mid-Market segment has highest outstanding amount at ₹46.6L
- Enterprise outstanding is highest at ₹1.62 Cr — concentrated risk
- Retail segment has fewest overdue invoices (23) despite slowest segment avg
- Enterprise pays fastest on average (26.7 days) but owes most when they don't pay

Insight

Enterprise customers create the highest financial risk when they default — not because they default most frequently, but because each invoice is so large. SME customers are the most consistently slow payers. Mid-Market segment represents the largest volume of problematic outstanding.

Business Action

Implement segment-specific credit policies: Enterprise — enhanced credit checks before contract signing, personal guarantee for invoices > ₹5L; SME — stricter 30-day terms with automated reminders from Day 20; Mid-Market — require 30% advance payment on new accounts.

Q7 — INVOICE DISPUTE ANALYSIS

Business Question:

Which invoice types get disputed most? What are the root causes?

Why This Matters:

Every dispute is a symptom of a process failure somewhere upstream — in pricing, delivery, contract management, or billing. Identifying patterns in dispute reasons allows the business to fix the root cause, not just the symptoms.

Query Result (PostgreSQL):

? Question 7: Which invoice types get disputed most?

```
SELECT invoice_category, dispute_reason, COUNT(*) AS dispute_count,
ROUND(SUM(invoice_amount), 2) AS disputed_value, ROUND(COUNT(*) * 100.0 / SUM(COUNT(*))
OVER(), 2) AS pct_of_disputes FROM invoices WHERE dispute_flag = 'Yes' GROUP BY
invoice_category, dispute_reason ORDER BY dispute_count DESC;
```

	invoice_category character varying (50)	dispute_reason character varying (100)	dispute_count bigint	disputed_value numeric	pct_of_disputes numeric
1	Consulting	Quality Issue	4	1679301.00	13.33
2	Consulting	Contract Disagreement	4	1728263.00	13.33
3	Consulting	Goods Not Received	3	24076.00	10.00
4	Annual License	Goods Not Received	3	88983.00	10.00
5	Maintenance Contract	Price Mismatch	3	91971.00	10.00
6	Annual License	Price Mismatch	2	171115.00	6.67
7	Annual License	Contract Disagreement	2	124794.00	6.67
8	Consulting	Price Mismatch	1	17624.00	3.33
9	Service Fee	Contract Disagreement	1	232861.00	3.33
10	Consulting	Duplicate Invoice	1	12856.00	3.33
11	Service Fee	Duplicate Invoice	1	32278.00	3.33
12	Product Sale	Duplicate Invoice	1	9173.00	3.33
13	Product Sale	Quality Issue	1	562404.00	3.33
14	Annual License	Duplicate Invoice	1	8006.00	3.33
15	Annual License	Quality Issue	1	158200.00	3.33
16	Maintenance Contract	Quality Issue	1	32374.00	3.33

Key Findings:

- ▶ Consulting invoices are most disputed — 4 cases each for Quality Issue and Contract Disagreement
- ▶ Annual License invoices have multiple disputes across Price Mismatch and Contract Disagreement
- ▶ Goods Not Received disputes appear in both Consulting and Annual License categories
- ▶ Quality Issue is the single most common dispute reason overall
- ▶ Service Fee and Product Sale also show dispute patterns around Duplicate Invoices

Insight

Consulting and Annual License are the highest-dispute categories. The recurrence of 'Quality Issue' and 'Contract Disagreement' in Consulting suggests that scope-of-work definitions are unclear at contract stage — leading to delivery expectation mismatches that translate into payment disputes.

Business Action

Mandate detailed Scope of Work (SOW) documentation for all Consulting engagements before invoice is raised. Implement a contract clarity checklist. For Annual License renewals, send a pricing confirmation email 30 days before invoice to eliminate Price Mismatch disputes. Create a dispute tracking dashboard for the Legal/Finance team.

Q8 — REALISTIC CASH RECOVERY — NEXT 30 DAYS

Business Question:

How much cash can we realistically recover in the next 30 days, and where should we focus?

Why This Matters:

The CFO needs a number — not a vague answer. This query provides a segment-by-segment, bucket-by-bucket recovery forecast with prioritisation, enabling the treasury team to plan cash flows with confidence.

Query Result (PostgreSQL):

	aging_bucket character varying (20)	customer_segment character varying (50)	invoice_count bigint	recoverable_amount numeric	recovery_action text
1	90+ Days	Enterprise	19	14119898.08	Critical - Write Off Review
2	90+ Days	Mid-Market	37	4048893.87	Critical - Write Off Review
3	61-90 Days	Enterprise	3	1820747.37	Low - Legal Notice Territory
4	90+ Days	SME	55	1230274.85	Critical - Write Off Review
5	61-90 Days	Mid-Market	3	358106.00	Low - Legal Notice Territory
6	31-60 Days	Enterprise	2	314493.87	Medium - Escalate to Manager
7	90+ Days	Retail	43	230515.11	Critical - Write Off Review
8	0-30 Days	Mid-Market	2	134107.08	High - Immediate Follow Up
9	31-60 Days	Mid-Market	1	124013.00	Medium - Escalate to Manager
10	61-90 Days	Retail	2	12523.04	Low - Legal Notice Territory
11	31-60 Days	SME	1	6772.78	Medium - Escalate to Manager
12	61-90 Days	SME	1	6667.92	Low - Legal Notice Territory
13	0-30 Days	Retail	1	1121.40	High - Immediate Follow Up

Key Findings:

- ▶ 90+ Days, Enterprise: ₹1.41 Cr (₹141.20L) across 19 invoices — largest single block, Critical/Write-Off Review
- ▶ 90+ Days, Mid-Market: ₹40.49L across 37 invoices
- ▶ 61-90 Days, Enterprise: ₹18.21L across 3 invoices — Legal Notice Territory
- ▶ 90+ Days, SME: ₹12.30L across 55 invoices — highest invoice count, lowest per-invoice value
- ▶ 90+ Days, Retail: ₹2.31L across 43 invoices — many small invoices, low total value
- ▶ 0-30 Days genuinely recoverable total: ₹1.35L (Mid-Market ₹1.34L + Retail ₹0.01L) — High/Immediate Follow-Up priority

Insight

The realistic 30-day cash recovery target is far smaller than first estimated — ₹1.4L, not ₹41.6L — because most of what looked like '0-30 day, easily recoverable' AR was actually misclassified Partial/Disputed invoices that were, in reality, much older. The corrected picture shows the business has very little genuinely fresh AR; 87.6% of outstanding value is already in the 90+ Day critical zone.

Business Action

Reset the 30-day collection target to a realistic ₹1.35L-₹5.8L range (0-30 and 31-60 buckets combined) rather than the previously assumed ₹38-42L. Treat the ₹1.41 Cr in Enterprise 90+ Day accounts as the single highest-priority write-off/legal review block — it alone is nearly 72% of the entire 90+ Day bucket. SME's high invoice count (55) at low individual value suggests a batch settlement or write-off approach would be more efficient than invoice-by-invoice collection effort.

Q9 — REGIONAL PAYMENT BEHAVIOUR

Business Question:

Which regions have the worst payment behaviour? Should credit policies differ by geography?

Why This Matters:

Payment culture varies significantly across Indian cities. Understanding regional patterns allows the business to apply differentiated credit policies — stricter terms where risk is higher, smoother terms where payment behaviour is strong.

Query Result (PostgreSQL):

? Question 9: Which regions have worst payment behaviour?

```

SELECT
  region,
  COUNT(*) AS total_invoices,
  ROUND(SUM(invoice_amount), 2) AS total_billed,
  ROUND(SUM(outstanding_amount), 2) AS total_outstanding,
  ROUND(SUM(outstanding_amount) * 100.0 /
    SUM(invoice_amount), 2) AS outstanding_pct,
  ROUND(AVG(days_overdue), 1) AS avg_days_overdue
FROM invoices
GROUP BY region
ORDER BY outstanding_pct DESC;

```

	region character varying (50)	totalInvoices bigint	total_billed numeric	total_outstanding numeric	outstanding_pct numeric	avg_days_overdue numeric
1	Kolkata	60	373215.00	139013.63	37.25	153.1
2	Ahmedabad	60	345308.00	105145.92	30.45	93.2
3	Pune	70	9326537.00	2754134.58	29.53	156.1
4	Mumbai	70	58486476.00	16255139.32	27.79	109.2
5	Hyderabad	60	1899229.00	516345.42	27.19	106.5
6	Delhi	60	7678446.00	1910985.37	24.89	89.1
7	Chennai	60	1841414.00	419818.71	22.80	52.6
8	Bangalore	60	1787684.00	307551.42	17.20	71.3

Key Findings:

- Kolkata has highest outstanding % at 37.25% — worst regional performance
- Ahmedabad second worst at 30.45% outstanding
- Pune at 29.53% outstanding with avg 156 days overdue — longest delays
- Mumbai: highest total billed (₹5.84 Cr) with 27.79% outstanding (₹1.62 Cr)
- Bangalore best performer at 17.20% outstanding — strongest payment culture

Insight

Kolkata and Ahmedabad show structurally weak payment behaviour — not just isolated incidents. Pune's 156-day average overdue is alarming. Bangalore stands out as the most reliable region. Mumbai's sheer volume makes its 27.79% outstanding figure a large absolute risk even at moderate percentage.

Business Action

Introduce region-specific credit policies: Kolkata and Ahmedabad — require advance payment or LC (Letter of Credit) for new orders above ₹1L; Pune — mandatory monthly payment plans for outstanding accounts; Bangalore — can offer extended 45-day terms as incentive for continued strong performance.

Q10 — COLLECTION EFFICIENCY RATIO

Business Question:

What is our collection efficiency ratio over time? Is the business collecting what it bills?

Why This Matters:

Collection Efficiency Ratio (CER) is the single most important AR KPI. It answers: for every ₹100 we bill, how much do we actually collect? A ratio consistently below 80% indicates a business in cash flow distress.

Query Result (PostgreSQL):

Question 10: Collection efficiency ratio by month

```

SELECT
  TO_CHAR(invoice_date, 'YYYY-MM') AS month,
  ROUND(SUM(amount_paid) * 100.0 /
    NULLIF(SUM(invoice_amount), 0), 2) AS collection_efficiency_pct,
  ROUND(SUM(invoice_amount), 2) AS total_billed,
  ROUND(SUM(amount_paid), 2) AS total_collected,
  ROUND(SUM(outstanding_amount), 2) AS total_outstanding
FROM invoices
GROUP BY TO_CHAR(invoice_date, 'YYYY-MM')
ORDER BY month;
  
```

	month text	collection_efficiency_pct numeric	total_billed numeric	total_collected numeric	total_outstanding numeric
1	2024-01	46.20	4249450.00	1963357.12	2286092.88
2	2024-02	72.62	2061076.00	1496670.31	564405.69
3	2024-03	95.90	5647623.00	5416335.26	231287.74
4	2024-04	74.88	4337505.00	3248081.12	1089423.88
5	2024-05	50.21	10203551.00	5123139.70	5080411.30
6	2024-06	68.67	4496861.00	3088185.50	1408675.50
7	2024-07	78.03	3044336.00	2375526.30	668809.70
8	2024-08	59.91	4360669.00	2612649.70	1748019.30
9	2024-09	69.73	4519463.00	3151615.91	1367847.09
10	2024-10	58.73	3966534.00	2329388.15	1637145.85
11	2024-11	84.64	5857634.00	4957724.78	899909.22
12	2024-12	81.52	4287664.00	3495469.05	792194.95
13	2025-01	92.76	3100167.00	2875611.31	224555.69
14	2025-02	59.51	3974489.00	2365242.76	1609246.24
15	2025-03	75.17	8937684.00	6718082.79	2219601.21
16	2025-04	87.80	3648347.00	3203067.35	445279.65
17	2025-05	97.32	5045256.00	4910027.52	135228.48

Key Findings:

- ▶ Best month: May 2025 at 97.32% — nearly perfect collection
- ▶ Worst month: Jan 2024 at 46.20% — less than half of billings collected
- ▶ Only 4 of 17 months achieved 90%+ collection efficiency
- ▶ March 2024 was strong at 95.90% — showing capability exists
- ▶ Most months cluster between 59–87% — below optimal 90% benchmark

Insight

The collection efficiency is highly volatile — swinging from 46% to 97% across 17 months. This volatility itself is a red flag: it suggests collections are event-driven (e.g., year-end pushes) rather than systematically managed. A well-run AR function should maintain 88–95% consistently.

Business Action

Establish a minimum monthly CER target of 85%. Introduce a Collections Dashboard that tracks real-time CER daily. Investigate January 2024 specifically — what caused the 46% collapse? Replicate whatever drove March 2024 (95.9%) and May 2025 (97.3%) as standard operating procedure.

6. EXECUTIVE SUMMARY & RECOMMENDATIONS

The AR Leakage Audit of 500 invoices spanning January 2024 to June 2025 reveals a business with strong billing activity but structurally weak collection discipline. Total outstanding across all non-paid categories amounts to ₹2.24 Crore — representing significant locked working capital that is directly impacting the company's cash position. A corrected aging calculation (see Section 5, Q2) further shows that 87.6% of this amount has been outstanding for 90+ days, a more severe concentration than initially estimated.

Key Performance Indicators

KPI	Value	Benchmark	Status
Total Invoices Analysed	500	—	✓
Total Billed (18 months)	~₹8.17 Cr	—	✓
Total Collected	~₹5.93 Cr	—	✓
Total Outstanding	~₹2.24 Cr	<10%	⚠
Collection Efficiency (Avg)	~72.6%	>90%	✗
Average DSO	~30.8 days	<30 days	⚠
90+ Day Overdue	₹1.96 Cr (87.6% of outstanding)	₹0	✗
Disputed & Blocked	₹49.7L	₹0	✗
Best Collection Month	May 2025 (97.3%)	>90%	✓
Worst Collection Month	Jan 2024 (46.2%)	>90%	✗

Top 5 Strategic Recommendations

1. Write-Off Provisioning Review for the 90+ Day Bucket

₹1.96 Cr — 87.6% of all outstanding AR — sits beyond 90 days overdue (154 invoices). This is now confirmed as the dominant problem, not a secondary one. Engage legal counsel and prepare a formal write-off provisioning review for the CFO immediately.

2. Realistic Collections Targeting

Genuinely fresh AR (0-30 and 31-60 day buckets combined) totals only ₹5.9L across 6 invoices. Set collection targets accordingly — the business should not expect a large near-term cash recovery from 'easy' buckets; most recoverable value requires sustained legal/negotiated settlement effort, not routine follow-up calls.

3. Sales-Finance AR Accountability Framework

Link sales commissions to cash collection. Introduce monthly joint AR review. Amit Verma and Priya Sharma's portfolios need urgent joint intervention.

4. Segment & Region-Specific Credit Policies

Kolkata, Ahmedabad, and Pune require stricter credit controls. SME segment needs automated reminders from Day 20. Enterprise accounts need personal guarantees for invoices above ₹5L.

5. Process Fix for Consulting Disputes

Mandate Scope of Work documentation before any Consulting invoice is raised. This single fix could eliminate 40%+ of current dispute volume.

Data correction note: these recommendations reflect a corrected Days_Overdue calculation. The original analysis underestimated aging severity because Partial and Disputed invoices were defaulting to 0 days overdue regardless of true age. The corrected figures show the AR problem is more concentrated in the 90+ day critical zone than first reported, and less recoverable through routine 30-day follow-up than originally estimated.

7. CFO ACTION MEMO

TO:	Chief Financial Officer
FROM:	Sumesh Achary, Data Analyst
DATE:	June 2026
SUBJECT:	AR Leakage Audit — Urgent Action Required on ₹2.24 Cr Outstanding
PRIORITY:	HIGH

A data-driven analysis of 500 invoices across 18 months has identified ₹2.24 Crore in outstanding receivables that require immediate management attention — of which 87.6% (₹1.96 Cr, fully detailed in Section 5 Q2) is concentrated in the 90+ day critical aging bucket. This memo summarises the three most urgent actions required.

URGENT ACTION 1 — Write-Off Provisioning Review

₹1.96 Cr of outstanding AR (154 invoices, 87.6% of the total) has been outstanding for 90+ days — a materially larger share than routine collections processes can be expected to recover. This must be treated as a write-off provisioning and legal review exercise, not a standard follow-up campaign. Recommend convening Finance and Legal within 5 business days to triage this bucket by recovery probability. Of this, Enterprise-segment accounts alone account for ₹1.41 Cr (nearly 72% of the 90+ Day bucket) and should be the first priority within this review.

URGENT ACTION 2 — Reset 30-Day Collection Expectations

Genuinely fresh AR (0-30 and 31-60 day buckets combined) totals only ₹5.8L across 7 invoices. Set realistic near-term collection targets against this smaller base rather than the broader outstanding figure — and treat any recovery from the 61-90 and 90+ buckets as a separate, longer-cycle workstream with its own tracking.

URGENT ACTION 3 — Dispute Resolution Task Force

₹49.7L is completely blocked in 30 disputed invoices. A cross-functional task force (Finance + Legal + Sales) must be convened within 5 business days to resolve each dispute. Consulting category disputes (primarily Quality Issue and Contract Disagreement) must be addressed with updated SOW documentation going forward.

This analysis was produced using PostgreSQL-based SQL queries on a structured AR dataset. Full query documentation, aging analysis, regional breakdown, and sales rep accountability data are available in the accompanying SQL file and Power BI dashboard.